Fiscal Year:	FY 2025	**Task Last Updated:**	FY 07/09/2025
PI Name:	Donahue, Henry J. Ph.D.		
Project Title:	Gap Junctions and Microgravity Induced Bone Loss		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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City:	Richmond	**State:**	VA
Zip Code:	23284-3067	**Congressional District:**	3
Comments:	Formerly at Pennsylvania State University, Hershey, PA, from 2012-2015.		
Project Type:	Ground	**Solicitation / Funding Source:**	2016-17 Space Biology (ROSBio) NNH16ZTT001N-MS, PS, AB. App D,E,F: Research Using Microgravity Simulation Devices, Parabolic and Suborbital Flights, and Antarctic Balloons
Start Date:	07/24/2018	**End Date:**	12/31/2023
No. of Post Docs:	2	**No. of PhD Degrees:**	
No. of PhD Candidates:	4	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA KSC
Contact Monitor:	Ruby, Anna Maria	**Contact Phone:**	321-867-7065
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 12/31/2023 per PI and NSSC information (Ed., 4/11/23) NOTE: End date changed to 12/31/2022 per NSSC information. (Ed. 8/25/22) NOTE: End date changed to 07/23/2022 per NSSC information. (Ed. 11/30/21)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Zhang, Yue Ph.D. (Virginia Commonwealth University)		
Grant/Contract No.:	80NSSC18K1473		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Astronauts exposed to extended periods of unloading experience significant bone loss. As NASA undertakes an exploration initiative involving long-duration spaceflight and exploration of asteroids and Mars, astronauts will encounter prolonged periods of unloading followed by subsequent extra-vehicular activities in a gravity environment. Loss of bone mass and strength will complicate these activities and impair mission efficiency. For instance, considering the magnitude of bone loss occurring during spaceflight (0.5-1.5% loss of bone mineral density (BMD)/month), a fall on an already weakened skeletal system could result in fracture. Therefore, development of countermeasures for microgravity-induced bone loss is critical. We previously demonstrated that osteoblast- and osteocyte-specific deficiency in the gap junctional protein connexin 43 (Cx43) is protective against bone loss induced by unloading via hindlimb suspension (HLS). However, the mechanism by which this occurs is unknown. Osteocytes are well situated in bone to detect and communicate mechanical signals to effector cells and therefore play a pivotal role in bone's ability to sense and respond to unloading from exposure to microgravity. Therefore, we will examine the hypothesis that Cx43 deficiency attenuates the response of osteocytes to unloading resulting from simulated microgravity. Additionally, we will examine whether the response of osteocytes exposed to simulated microgravity in vitro can predict bone response to simulated microgravity in vivo. Using an in vivo and in vitro model, we will complete the following two aims. Aim 1: Examine the response of wild-type and Cx43-deficient (via CRISPR gene editing) murine osteocytes (OCY454) to simulated microgravity in vitro using a High Aspect Ratio Vessel (HARV). Aim 2: Examine the effect of simulated microgravity in vivo on bones from osteocyte-specific Cx43-deficient mice. Cx43 will be selectively deleted from mice using a Cre recombinase approach similar to what we previously used to delete Cx43 from osteoblasts and osteocytes. Using well-established experimental paradigms like the HARV apparatus and HLS greatly increases the probability of successfully completing our aims and attaining important, novel findings. Collectively, our infrastructure and expertise, in combination with multiple unique reagents, place us in an unparalleled position to rapidly and significantly advance knowledge pertaining to the mechanisms underlying microgravity-induced bone loss. These aims and the experiments designed therein represent a substantive departure from our current understanding and will reveal novel targets for countermeasures.
Rationale for HRP Directed Research:	
	The proposed research will establish the mechanism underlying the protective effect of Cx43 against microgravity-induced bone loss. This will help fill gaps in our understanding of gravity-sensing mechanisms. We will also gain insight as to whether in vitro models of microgravity, such as culturing cells in a HARV apparatus, induce similar changes in osteocytic gene expression as do in vivo models such as HLS. From the perspective of microgravity countermeasures, this information will guide the choice of potential therapeutics. On the ground, concomitant bone loss is frequently experienced by individuals confined to bed for long durations, whether as a result of neurological injury, trauma, or neuromuscular disease (e.g., cerebral palsy and muscular dystrophy). In these patients, and in healthy patients subjected to extended bed rest, a significant loss of bone occurs. In the case of recoverable disease or when ambulation is still possible, this leaves patients at risk for falls and catastrophic fracture. Understanding the mechanisms involved in bone loss during spaceflight, and the development of countermeasure strategies, will provide a framework for the development of preventative treatments for astronauts and the clinical population.
Research Impact/Earth Benefits:	The proposed research is innovative because it provides the first simultaneous examination of effects of simulated microgravity on genetically modified osteocytes in vivo and in vitro. A few studies have examined the effects of simulated microgravity on bone in vivo and in vitro,but no study has compared the effects of simulated microgravity on osteocytes or genetically modified mice in vivo and in vitro in the same study. Osteocytes play a key role in bone's mechanosensitivity and adaptation to changes in mechanical loading. Furthermore, we are investigating the novel concept that Cx43 deficiency protects against microgravity-induced bone loss through a non-canonical mechanism involving an interaction between Cx43 and the WNT/ß-catenin pathway. Finally, we will use innovative transcriptomic approaches to compare the response of primary osteocytes exposed to simulated microgravity in vivo to osteocytic OCY454 cells exposed to simulated microgravity in vitro. In this proposal we are leveraging our experience and expertise in bone biology (Donahue) and gene expression (Buck). Our team brings forward a collaborative and unique skill set in order to ensure a complete and comprehensive investigation of the response of osteocytic cells to simulated microgravity.
Task Progress:	NOTE: No final report received; however, publications have been added to update this report in the NASA Task Book (Ed., 7/9/2025).
Bibliography Type:	Description: (Last Updated: 01/06/2026)
Articles in Peer-reviewed Journals	Zeineddine Y, Friedman MA, Buettmann EG, Abraham LB, Hoppock GA, Donahue HJ. "Genetic diversity modulates the physical and transcriptomic response of skeletal muscle to simulated microgravity in male mice." npj Microgravity. 2023 Dec 1;9:86. https://doi.org/10.1038/s41526-023-00334-8 ; PMID: 38040743; PMCID: PMC10692100 , Dec-2023
Articles in Peer-reviewed Journals	Meas SJ, Daire GM, Friedman MA, DeNapoli R, Ghosh P, Farr JN, Donahue HJ. "A comparison of bone microarchitectural and transcriptomic changes in murine long bones in response to hindlimb unloading and aging." Bone. 2024 Feb:179:116973. https://doi.org/10.1016/j.bone.2023.116973 ; PMID: 37996046; PMCID: PMC11651238 , Feb-2024
Articles in Peer-reviewed Journals	Hoppock GA, Buettmann EG, Denisco JA, Goldscheitter GM, Condyles SN, Juhl OJ 4th, Friedman MA, Zhang Y, Donahue HJ. "Connexin 43 and cell culture substrate differentially regulate OCY454 osteocytic differentiation and signaling to primary bone cells." Am J Physiol Cell Physiol. 2023 Oct 1;325(4):C907-C920. https://doi.org/10.1152/ajpcell.00220.2023 ; PMID: 37602413 PMCID: PMC10635658 , Oct-2023

Articles in Peer-reviewed Journals	Friedman MA, Buettmann EG, Zeineddine Y, Abraham LB, Hoppock GA, Meas SJ, Zhang Y, Farber CR, Donahue HJ. "Genetic variation influences the skeletal response to hindlimb unloading in the eight founder strains of the diversity outbred mouse population." J Orthop Res. 2024 Jan;42(1):134-140. https://doi.org/10.1002/jor.25646 ; PubMed PMID: 37321985; PubMed Central PMCID: PMC10721729 , Jan-2024
Articles in Peer-reviewed Journals	Maroni CR, Friedman MA, Zhang Y, McClure MJ, Fulle S, Farber CR, Donahue HJ. "Genetic variability affects the response of skeletal muscle to disuse." Journal of Musculoskeletal & Neuronal Interactions. 2021;21(3):387. PMCID: PMC8426660; PMID: 34465678 , Dec-2021
Articles in Peer-reviewed Journals	Meas SJ, Daire GM, Buettmann EG, Donahue HJ. "Hindlimb unloading results in accelerated bone loss compared to aging in select bone compartments of female murine femurs." J. Biomech. 2026 Jan;195:113134. https://doi.org/10.1016/j.jbiomech.2025.113134 ; PMID: 41435464 , Jan-2026